

Deep Reinforcement Learning Course Project

2026 Spring

April 2026

0 Overview

This is the documentation for the **course project of Deep Reinforcement Learning in 2026 Spring**. You are required to finish a group project related to some sub-field of deep reinforcement learning. We have provided some tracks for you to explore, and we are also open to your own super cool ideas beyond our proposed tracks (you can choose the Open Track if you want to work on your own idea).

In this documentation, we will first explain our logistics ([Section 1](#)), resources ([Section 2](#)) and timeline ([Section 3](#)), and **the requirements of different research tracks are attached at the end of this documentation**.

Good luck and enjoy your experience in deep reinforcement learning research! :)

1 Logistics

1.1 Rules

- This is a group project and each group consists of **2-5 students**. Please fill in this [google sheet](#) after you have found your group. **No later than May 5th!**
- You will need to submit a 1-page project proposal, some milestone materials, and a report to summarize your project (3-page minimum and 8-page maximum).
- This project weighs 40% in the whole score of this course. **The project proposal, oral/poster session and the report occupy 5%, 17% and 18% respectively.**
- The milestone materials, which can be of any format (e.g., report/demo video/live demo/website/slides) that can demonstrate your results, are only used for selecting oral presentations. You should submit your milestone material some time before the presentation day (e.g., Week 15).
- Poster and oral sessions are expected to be in English. Requirements on the presentation's duration will be announced later.
- We expect to host the presentation day sometime near the end of the semester, before the exam weeks (e.g., Week 16).
- We expect each group member to contribute to the project genuinely; contributions should be listed clearly.

1.2 Report Format

Please use the provided [ICLR 2026 LaTeX template](#) to complete your report.

Your report should include the following sections:

- A short abstract that summarizes your work;
- An introduction section that explains your motivation, proposed method, and results;
- A related work section that briefly surveys the past literature;
- A method section that describes how you solve the problem in detail;
- A results section that demonstrates the performance of your proposed method;
- (Optional) Additional ablative experiments to understand your proposed method;
- (Optional) A discussion section to elaborate on your thoughts on your project.

Here is a detailed list of how we will evaluate your report:

- Writing, format, and style of your report: 10%;
- Literature review: 10%;
- Novelty (if the project is your own idea) or depth that you investigate the idea (if you select one of the ideas we provided): 20%. You can also make your own variation over the idea we provided, we will evaluate this term in a case-by-case fashion;
- Results, how well does your solution solve the task, and how impressive are the final results: 50%;
- Logical clarity, how well your method and your results are delivered, and whether it's clear to draw the conclusion you provided from your results: 10%;

1.3 Poster Format

Your poster should include the following sections:

- A short abstract that summarizes your work;
- A motivation section that briefly explains why you want to work on this idea;
- A problem formulation section to introduce your task setting in an RL context;
- A method section to demonstrate your key improvement/contribution over existing methods/dataset/simulator, etc;
- A result section that shows your main experiment figures, curves and tables.

Here is a detailed list of how we will evaluate your poster:

- Visual Appearance of your poster: 30%;
- Clarity of your method and results: 30%;
- Faithfulness and integrity compared to your report: 30%;
- Guidance or inspiration for future research: 10%.

You can design your poster through LaTeX ([template](#)), PowerPoint ([template](#)), or any image editing tool you find convenient. Only the final figure (in png, jpg, pdf, etc.) is required. **The aspect ratio of your poster should be 16:9.** Those selected for oral don't need to provide a poster.

1.4 Oral Format

Your oral presentation should include the following sections:

- A motivation section that briefly explains why you want to work on this idea;
- A problem formulation section to introduce your task setting in an RL context;
- A method section to demonstrate your key improvement/contribution over existing methods/dataset/simulator, etc;
- A result section that shows your main experiment figures, curves and tables.

Here is a detailed list of how we will evaluate your presentation:

- Clarity of your oral presentation: 30%;
- Visual appearance of your slides: 20%;
- Literature review: 10%;
- Presentation of methods and results: 30%;
- Guidance or inspiration for future research: 10%.

Note: Results are best viewed with **comparison videos**, e.g., your RL agent successfully wins a game while others cannot.

1.5 Tips

- Properly identify the scope of your research problem. Something like “Applications of Deep RL” is too broad, and “PPO speedup on a 3090 Ti GPU” is probably too narrow;
- Finding a small number of high-quality reference papers will save you a lot of time. It is a good idea to start with top-tier journals or conference papers with high citations;
- Start from a working/high-star repo in GitHub. RL is fragile; always start with something that already works;
- Rehearse your talk!
- If your project is super cool, please consider extending it and submitting it to an AI conference.

2 Resources

2.1 RL Algorithms Implementation

- [Stable Baselines3](#)
- [Tianshou](#)
- [ElegantRL](#)
- [Spinningup](#)
- [RLlib](#)
- [Dopamine](#)

2.2 Popular RL Simulators & Environments

Simulators:

- [MuJoCo](#)
- [PyBullet](#)
- [Isaac Gym](#)
- [Sapien](#)
- [Gazebo](#)
- [SMAC](#)
- [PlasticineLab](#)
- [TaiChi Lang](#)

Environments:

- [Gymnasium\(Openai Gym\)](#)
- [DeepMind Control Suite](#)
- [Meta-World](#)
- [IsaacGymEnvs](#)
- [RLBench](#)
- [robosuite](#)
- [HumanoidBench](#)

2.3 Useful tools:

- [WandB](#)
- [TensorBoard](#)
- [Hydra](#)

2.4 Reference Resources

- [arXiv](#): search for relevant categories or keywords
- [Google Scholar](#): search for relevant keywords or authors
- [ICML/ICLR](#): proceedings of top AI conferences
- Websites of research labs or organizations, such as [OpenAI](#), [Google DeepMind](#), etc.

3 Timeline

Week 10:

- Project starts!
- Documentation released
- Team up (fill in the [google sheet](#) **no later than May 5th, Tuesday of Week 11**; you are not required to decide your track at that time)

Week 12:

- 1-page project proposal
- Idea selected or idea proposed
- Team members and your track
- If you propose your own idea: idea justification.
- Key literature survey (list 3 key papers about your selected idea)

Week 15:

- Mid of this week (Wednesday): Milestone materials due
- End of this week (Friday): Notification of oral and poster

Week 16:

- First 45 mins: Oral presentations
- Second 45 mins: Poster session

After Week 16:

- Final report due (refer to Section [1.2](#) for requirements)
- Deadline will be announced later (will try to avoid the final exam weeks).

Dexterous Piano Track

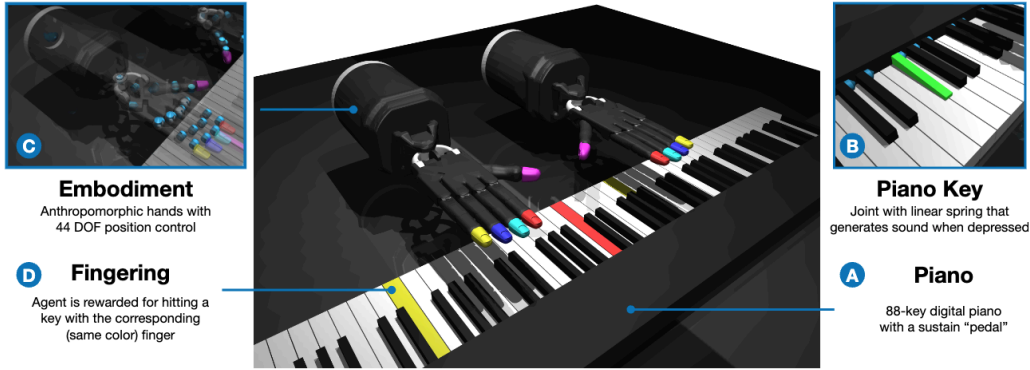


Figure 1: **ROBOPIANIST** simulation featuring a full-size digital keyboard (A) with 88 piano keys modeled as linear springs (B). In the piano playing task, two (left and right) anthropomorphic Shadow hands (C) are tasked with playing a musical piece encoded as a trajectory of key presses (D).

1 Project Description

This track requires you to investigate how to teach a robot to play the piano. Recent works have shown significant breakthroughs in robot skill levels across various locomotion and manipulation tasks. One of the skills that best showcases human dexterity is playing the piano. Equipping robots with human-level dexterity will be an important topic in future robot learning.

2 Requirements

In this section, we will provide a reference technology roadmap for your project. Note that you are not required to finish all the steps in this roadmap.

2.1 Understanding Basic Concepts and Algorithms

Provide a comprehensive literature review on robot playing piano, identify current challenges, and give a brief introduction to the technical details of Robopianist [PI1], PianoMime [PI2], and RP1M [PI3].

2.2 Running Baselines

You are required to conduct experiments on the original repo of [PianoMime](#). You should select **3** songs from the training dataset that you believe sound the best for the robot to play. You'll also need to visualize the F1 score training curve and produce a final video of the performance.

Meanwhile, you should also train a multi-song policy (or use the given checkpoints in repo), and show the performance video in unseen songs (i.e., in test dataset).

3 Algorithm Design

You need to improve both the F1 score of the single-song policy and the generalist policy. For the single-song policy, you can choose a piece from the training dataset that you consider relatively difficult and longer in duration, then compare the performance of your improved approach against the baseline and provide both the performance curve and a play-through video. For the generalist policy, you need to report both the improved and baseline F1 scores, and show a performance comparison on 5 pieces.

Try to make your robot perform the piece smoothly with behavior that closely resembles real-world conditions!

4 Potential Directions

If you are interested in this project but have no idea on where to start, might as well have a try on these directions:

- **Design better reward function:** For single-song policy, The reward function is the foundation for an RL policy's learning. During the performance, it can be observed that the fingers sometimes shake violently, and the overall hand movement can be quite extreme, which differs significantly from real-world conditions. Therefore, we can design a better reward function to alleviate this issue.
- **Integrate novel imitation learning tricks:** For the generalist policy, you can leverage any novel IL method to improve both the high-level and low-level aspects of its performance.
- **Implement your own goal representation:** For the generalist policy, the quality of the goal representation is a critical factor in determining whether the task can be completed. You can improve the quality of this representation by leveraging representation-learning techniques to further enhance performance.
- **Leverage RL tricks:** For the single-song policy, you can apply any RL strategies to enhance the robot's performance.

4.1 Exploration

You can consider mounting both Shadow Hands on the [UR10](#) robot arms instead of letting them float in midair, and compare the performance before and after installing the arm.

5 About Grading

We provide a reference grading policy for this project as follows:

- **[60 pts]:** Report the baseline results, and try a trick.
- **[80 pts]:**
 - Report the baseline results, conduct thorough experiments on your proposed tricks, and analyze the results in detail (it can be okay if all of them fail).
 - Report the baseline results, propose and implement your own idea, and analyze the results in detail (it can be okay if it fails).
- **[100 pts]:** Make a great improvement on PianoMime and perform pieces smoothly.
- **Bonus:** Achieve comparable performance when equipped with arms.

6 Contact

If you have any questions, please feel free to contact our TAs via WeChat or email:

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- Maanping Shao: smap24@mails.tsinghua.edu.cn
- Junhua Yao: junhuayao41@gmail.com

Dexterous Piano Track Reference

- [PI1] Kevin Zakka, Philipp Wu, Laura Smith, Nimrod Gileadi, Taylor Howell, Xue Bin Peng, Sumeet Singh, Yuval Tassa, Pete Florence, Andy Zeng, et al. Robopianist: Dexterous piano playing with deep reinforcement learning. *arXiv preprint arXiv:2304.04150*, 2023.
- [PI2] Cheng Qian, Julen Urain, Kevin Zakka, and Jan Peters. Pianomime: Learning a generalist, dexterous piano player from internet demonstrations. *arXiv preprint arXiv:2407.18178*, 2024.
- [PI3] Yi Zhao, Le Chen, Jan Schneider, Quankai Gao, Juho Kannala, Bernhard Schölkopf, Joni Pajarinen, and Dieter Buehler. Rp1m: A large-scale motion dataset for piano playing with bi-manual dexterous robot hands. *arXiv preprint arXiv:2408.11048*, 2024.

Quadruped Parkour Track



Figure 2: **Robot Parkour Learning (RPL)** proposes a system for learning a single end-to-end vision-based parkour policy of diverse parkour skills using a simple reward without reference motion data. It develops a reinforcement learning method to generate parkour skills, including climbing, leaping, crawling, squeezing, and running. These skills are then distilled into a single vision-based parkour policy and transferred to a quadrupedal robot.

1 Project Description

Parkour (in Fig.2) is a grand challenge for legged locomotion that requires robots to overcome various obstacles rapidly in complex environments. Furthermore, autonomous parkour requires robots to learn generalizable skills that are both vision-based and diverse to perceive and react to various scenarios.

Your team should first conduct a thorough literature review on this field, and understand how the existing approaches tackle this challenging locomotion problem. Additionally, you should read the paper of some state-of-the-art algorithms [QP1, QP2, QP3] and the code base of [Robot Parkour Learning](#) thoroughly. Then you should reproduce the results from the SOTA methods, transfer the policy to another platform to test generalization and propose your own techniques to improve the performance.

Notice: This project involves a GPU-based simulator named Issac Gym. It may be unsuitable for you if you only have a Windows or Mac laptop. Moreover, a local Linux machine with a display is also recommended.

2 Requirements

In this section, we will provide a reference technology roadmap for your project.

2.1 Understanding Basic Concepts and Algorithms

Provide a comprehensive literature review on parkour learning, identify current challenges, and give a brief introduction to the technical details of RPL [QP2].

2.2 Reproducing Results from RPL

First of all, you are required to reproduce the results with the code base of [RPL](#) on the Unitree Go2 robot in Issac Gym. The method involves two parts, in which the agent first learns separate parkour skills with a two-stage RL algorithm and then distills these skill policies into a single visual policy. You need to provide videos and training details for both stages.

2.3 Sim2sim Transfer

Since it is hard to provide real robots to test the performance of the distillation policy, you are alternatively required to transfer the policies in Issac Gym to a new environment in the [MuJoCo](#) simulator. Unlike Issac Gym which is designed for accelerating RL with GPU utilization, MuJoCo is a physics simulation engine that has more precise physical modeling and smaller dynamic gaps. In this part, you should construct parkour environments with different difficulty levels for the Unitree Go2 robot by yourself, and deploy the trained distillation policy. You need to provide videos of your distillation policy in parkour environments with different difficulties.

You may find the following repos helpful to your implementation: [mujoco menagerie](#), [unitree rl gym](#), [lucidsim](#).

2.4 New Improvements

You are also encouraged to propose improvements from any perspective based on RPL. You can even try submit your project to a top conference if you make great contributions!

2.5 Potential Directions

If you are interested in this project but have no idea on where to start your improvements, might as well have a try on these directions:

- **Conquer more challenging environments and tasks:** You can try to design new methods and skills to learn and distill more challenging tasks, including running faster, leaping further, or parkouring in more complex environments.
- **Use LLM-generated rewards to automate RL:** Rewards in RL usually require careful manual designs, especially for complex skills. However, with the rapid development of LLM, some works [QP4, QP5] try to automatically generate and test rewards with LLM such as GPT-4. You can try to design a new pipeline to generate some new skills other than the five proposed in RPL paper.
- **Make use of more modalities of observations:** To reduce visual gaps between simulation and real-world, RPL only uses depth information as observation, which may hinder policy performance in the deployment environments. You can try to make use of more observation information to further boost the parkour ability of robots. Some common tricks can be data augmentation, domain randomization, and real-to-sim environment transfer [QP3].

3 About Grading

We provide a reference grading policy for this project as follows:

- **[60 pts]**: Report the reproduction results, including materials for both RL training and distillation stages.
- **[70 pts]**: Construct a MuJoCo environment for sim2sim transfer.
- **[80 pts]**: Complete sim2sim transfer of the distillation policy.
- **[80-100 pts]**: Make some improvements on RPL. The scores may vary according to the contributions of your improvements.

4 Contact

If you have any questions, please feel free to contact our TAs via WeChat or email:

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- Maanping Shao: smap24@mails.tsinghua.edu.cn
- Junhua Yao: junhuayao41@gmail.com

Quadruped Parkour Reference

- [QP1] Xuxin Cheng, Kexin Shi, Ananye Agarwal, and Deepak Pathak. Extreme parkour with legged robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11443–11450. IEEE, 2024.
- [QP2] Ziwen Zhuang, Zipeng Fu, Jianren Wang, Christopher Atkeson, Soeren Schwertfeger, Chelsea Finn, and Hang Zhao. Robot parkour learning. *arXiv preprint arXiv:2309.05665*, 2023.
- [QP3] Alan Yu, Ge Yang, Ran Choi, Yajvan Ravan, John Leonard, and Phillip Isola. Learning visual parkour from generated images. In *8th Annual Conference on Robot Learning*, 2024.
- [QP4] Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Eureka: Human-level reward design via coding large language models. *arXiv preprint arXiv:2310.12931*, 2023.
- [QP5] Yecheng Jason Ma, William Liang, Hung-Ju Wang, Sam Wang, Yuke Zhu, Linxi Fan, Osbert Bastani, and Dinesh Jayaraman. Dreureka: Language model guided sim-to-real transfer. *arXiv preprint arXiv:2406.01967*, 2024.

Humanoid Dancer Track



Figure 3: **Physics-based motion imitation** leverages physics simulation and reinforcement learning to teach characters a diverse range of motion skills—such as walking, running, jumping, and rolling—directly from motion capture data. The process involves collecting high-quality mocap data, retargeting it to the character’s skeleton, training a control policy via reinforcement learning in a physics-based environment, and finally deploying and testing the policy to ensure robust and realistic performance in complex scenarios.

1 Project Description

Physics-based motion imitation combines physics simulation and deep learning to enable virtual characters or robots to replicate reference motions, often from motion capture data. Modern approaches mainly use reinforcement learning (e.g., PPO, SAC) to train control policies that ensure both visual accuracy and physical feasibility. This technique is widely applied in realistic animation and robot motion control—enhancing naturalness in animation and enabling human-like behavior in robotics. This track requires to understand key components of an existing codebase, including data processing, reward design, and policy training, and to explore improvements in functionality or performance.

Notice: This project involves a GPU-based simulator named Isaac Gym. It may be unsuitable if you only have a Windows or Mac laptop. Moreover, a local Linux machine with a display is also recommended.

2 Requirements

In this section, we will provide a reference technology roadmap for your project.

2.1 Understanding Basic Concepts and Algorithms

Provide a comprehensive literature review on physics-based motion imitation learning. Identify current framework designs and challenges. Give a brief introduction to the technical details of mainstream methods such as DeepMimic [HD1], AMP [HD2], ASE [HD3], PHC [HD4], ExBody [HD5], H2O [HD6], HumanoidPlus [HD7].

2.2 Running Baselines

You are required to conduct experiments using the provided repository, [Humanoid Dancer](#).

The repository utilizes the H1 retargeted motion dataset from H2O [HD6]. It is based on the AMASS dataset, which provides physically compatible reference motions for training.

The policy training follows the DeepMimic paradigm, modeling the task as a POMDP. It employs goal-conditioned reinforcement learning using PPO. The reward function consists of two parts: regularization terms that constrain joint angles, fix gait, and ensure physical plausibility; and exponential tracking rewards that promote accurate imitation of the target motion.

Running the baseline implementation yields a control policy that can approximately follow the reference motion. However, the resulting behavior typically suffers from low tracking accuracy, unstable dynamics, and unnatural gait patterns.

2.3 Making Improvements

The objective of this final project component is to improve the performance and realism of the humanoid dancer policy. Your goal is to enhance motion quality, tracking accuracy, and the stability of the learned behavior. The improved system should demonstrate more natural and robust humanoid motion compared to the provided baseline.

2.4 Potential Directions

If you are looking for directions to improve the baseline system, here are a few potential areas to explore:

- **Provide more comprehensive reference motion data and fine-tune reward function coefficients:** The current implementation primarily focuses on joint position tracking, and the reward function coefficients have not been carefully adjusted. You can improve the reward design by including additional reference motion information and refining the reward weights to better match the desired behavior.
- **Utilize more advanced learning algorithms:** The baseline currently uses a simple PPO algorithm. You could explore advanced motion learning algorithms, such as AMP [HD2] or other state-of-the-art methods, to improve learning efficiency and policy performance.
- **Apply curriculum learning:** To facilitate learning of complex tasks, you can experiment with curriculum learning techniques. An approach could be to categorize the movements in the data set into different difficulty levels [HD4] or implement a reward curriculum [HD6] to ease the learning process.

3 About Grading

We provide a reference grading policy for this project as follows:

- **[60 pts]:**
 - Report the baseline results, including quantitative metrics such as success rate, tracking accuracy, and stability of the current control policy.
 - Record and include videos of the motion execution, demonstrating the current performance of the baseline.
 - Implement a modification to the current framework and provide a detailed analysis of the modification.
- **[80 pts]:**

- Improve the baseline results under certain constraints or conditions using specific methods or tricks. For example, you can select a subset of motion data to create a new dataset or modify existing motion sequences.
- Demonstrate some improvement in the baseline results in terms of both quantitative metrics and visual performance. This could include achieving a more human-like gait or enabling the humanoid to perform a variety of in-place actions, such as squatting or standing up.
- **[100 pts]:**
 - Achieve significant improvement in both quantitative metrics and visual performance compared to the baseline. The humanoid should be able to replicate most of the motions in the dataset and perform more expressive and dynamic actions such as long jumps or running.
 - Collect dance data outside the provided dataset, using sources like online videos or other methods, and successfully reproduce those motions using your methods.
- **Bonus:** Successfully replicate your method on a different humanoid robot.

4 Contact

If you have any questions, please feel free to contact our TAs via WeChat or email:

- Tianhai Liang: liangth24@mails.tsinghua.edu.cn
- Maanping Shao: smap24@mails.tsinghua.edu.cn
- Junhua Yao: junhuayao41@gmail.com

Humanoid Dancer Reference

- [HD1] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. Deepmimic: Example-guided deep reinforcement learning of physics-based character skills. *ACM Trans. Graph.*, 37(4):143:1–143:14, July 2018.
- [HD2] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. Amp: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics (ToG)*, 40(4):1–20, 2021.
- [HD3] Xue Bin Peng, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. Ase: Large-scale reusable adversarial skill embeddings for physically simulated characters. *ACM Trans. Graph.*, 41(4), July 2022.
- [HD4] Zhengyi Luo, Jinkun Cao, Alexander Winkler, Kris Kitani, and Weipeng Xu. Perpetual humanoid control for real-time simulated avatars. *arXiv preprint arXiv:2305.06456*, 2023.
- [HD5] Xuxin Cheng, Yandong Ji, Junming Chen, Ruihan Yang, Ge Yang, and Xiaolong Wang. Expressive whole-body control for humanoid robots, 2024.
- [HD6] Tairan He, Zhengyi Luo, Wenli Xiao, Chong Zhang, Kris Kitani, Changliu Liu, and Guanya Shi. Learning human-to-humanoid real-time whole-body teleoperation. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 8944–8951. IEEE, 2024.

- [HD7] Zipeng Fu, Qingqing Zhao, Qi Wu, Gordon Wetzstein, and Chelsea Finn. Humanplus: Humanoid shadowing and imitation from humans. *arXiv preprint arXiv:2406.10454*, 2024.

Open Track

1 Project Description

You have the option to choose your own topic in the course. For this project, You must follow these tips:

- Your proposed project should be related to our course.
- **You must discuss your proposal with TAs and get their approval, no matter whether you are proposing your brand new idea or just working on the project in your own lab.**
- **You cannot use the papers that you have already posted on arXiv or have published.**

2 Requirements

In this section, we will provide a reference technology roadmap for your project. Note that it is just a reference and we can help you to build and adjust your roadmap based on your specific idea.

2.1 Basic concepts and algorithms

Provide a comprehensive review of the literature on this field or the related work of your own solution.

2.2 Run Baselines

Run the baselines in your setting.

2.3 Algorithm Design

Design your own algorithm.

3 About Grading

We will discuss with you on a reference grading policy after you decide to work on your own project and get our approval.

4 Contact

If you choose this track, please first contact our TAs via WeChat or email, and we will schedule a discussion with you on your idea.